

# Algorithmic Pseudocode Sample 16

Source-backed Image2Struct algorithm sample

## 1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

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**Algorithm 1** Source-backed algorithmic procedure

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**Require:**  $n, p \in \mathbb{N}, \omega \in [0, 1], x \in \mathbb{R}^n, v \in \mathbb{R}^n$ .

- 1: Find a sorting permutation  $\sigma$  of vector  $x$ .
  - 2: Apply the  $\sigma$  to  $x$  and  $v$ , in place.
  - 3: **for**  $\tilde{\omega} = 1, 0$  **do**
  - 4:   **if**  $\tilde{\omega} = 0$  **then** ▷ Shift  $v$  by one place because of 0s on diagonal of  $\mathbf{I}$ .
  - 5:     **for**  $i = n, n - 1, \dots, 2$  **do**
  - 6:        $v_i \leftarrow v_{i-1}$
  - 7:     **end for**
  - 8:      $v_1 \leftarrow 0$
  - 9:   **end if**
  - 10: Identify the set of groups  $G$  of elements of  $v$  corresponding to equal entries in  $x$ .
  - 11:  $\tilde{v} \leftarrow v$
  - 12: **for**  $g$  in  $G$  **do**
  - 13:   Calculate sum  $\mathbf{s}$  of  $v$ 's elements in group  $g$ .
  - 14:   Set 0 in all  $\tilde{v}$ 's elements in group  $g$ .
  - 15:   **if**  $\tilde{\omega} = 0$  **then**
  - 16:     Set the  $\tilde{v}$ 's first element in group  $g$  to  $\mathbf{s}$ .
  - 17:   **else**
  - 18:     Set the  $\tilde{v}$ 's last element in group  $g$  to  $\mathbf{s}$ .
  - 19:   **end if**
  - 20: **end for**
  - 21:  $c_{\tilde{\omega}} \leftarrow$  cumulative sum of  $\tilde{v}$
  - 22: **end for**
  - 23:  $c \leftarrow \omega c_1 + (1 - \omega)c_0$  ▷ Multiplication and addition element-wise.
  - 24: Find the inverse of  $\sigma$ .
  - 25: Apply the  $\sigma^{-1}$  to  $c$ .
  - 26: **return**  $c$
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